

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12406675
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Sharifi; Matthew et al.

Multi-user warm words

Abstract

A method includes detecting a presence of multiple users within an environment of an assistant-enabled device (AED) and obtaining, for each user, a respective active set of warm words that each specify a respective action for a digital assistant to perform. Based on each respective active set of warm words, the method also includes executing a warm word arbitration routine to enable a final set of warm words for detection by the AED. Here, the final set of warm words includes warm words selected from the respective active set of warm words. While the final set of warm words are enabled, the method also includes receiving audio data corresponding to an utterance captured by the AED, detecting a warm word from the final set of warm words in the audio data, and instructing the digital assistant to perform the respective action specified by the detected warm word.

Inventors:	Sharifi; Matthew (Kilchberg, CH), Carbune; Victor (Zürich, CH)
Applicant:	Google LLC (Mountain View, CA)
Family ID:	88965364
Assignee:	Google LLC (Mountain View, CA)
Appl. No.:	18/056697
Filed:	November 17, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240169995 A1	May. 23, 2024

Publication Classification

Int. Cl.: G10L17/00 (20130101); G06F3/16 (20060101); G10L15/00 (20130101); G10L15/08 (20060101); G10L15/16 (20060101); G10L15/22 (20060101); G10L17/22 (20130101)

U.S. Cl.:

CPC **G10L17/22** (20130101); **G06F3/167** (20130101); **G10L15/22** (20130101); G10L15/00 (20130101); G10L2015/088 (20130101); G10L15/16 (20130101); G10L2015/223 (20130101); G10L17/00 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9691384	12/2016	Wang	N/A	G10L 15/18
10931999	12/2020	Jobanputra et al.	N/A	N/A
11132989	12/2020	Tolomei	N/A	H04R 3/005
11468381	12/2021	Harris	N/A	G06F 16/9035
11676595	12/2022	Igarashi	704/251	H04N 21/4722
11741969	12/2022	Rao	726/28	H04L 12/2829
11948563	12/2023	Liu	N/A	G06Q 10/1095
12218950	12/2024	Wang	N/A	H04L 63/20
2015/0007046	12/2014	Branson	709/217	H04L 67/1025
2016/0085565	12/2015	Arcese	726/7	H04L 63/0861
2016/0299661	12/2015	Alpert	N/A	G06T 17/05
2018/0130469	12/2017	Gruenstein	N/A	G10L 15/30
2018/0182390	12/2017	Hughes	N/A	G10L 15/22
2018/0323991	12/2017	Segal	N/A	G10L 15/26
2019/0012139	12/2018	Han	N/A	G10L 15/05
2019/0189019	12/2018	Hammersley	N/A	G06N 20/00
2020/0012906	12/2019	Albadawi	N/A	G06V 40/173
2020/0134211	12/2019	Miller	N/A	H04L 9/088
2020/0211563	12/2019	Arunachalam	N/A	G10L 15/30
2020/0342855	12/2019	Casado	N/A	G10L 15/00
2021/0134280	12/2020	Kurtz	N/A	G10L 15/30
2021/0201916	12/2020	Touret	N/A	H04M 3/42102
2021/0337307	12/2020	Wexler	N/A	G06V 40/161
2021/0366466	12/2020	de La Jonquière	N/A	G06N 3/08
2021/0375267	12/2020	Patil	N/A	G10L 15/07
2022/0139388	12/2021	Sharifi	704/231	H04M 3/40
2022/0157305	12/2021	Sakurai	N/A	G05D 1/223
2022/0171800	12/2021	Kumaresan	N/A	G06F 11/321
2022/0180862	12/2021	Sharifi	N/A	G10L 15/04
2022/0189465	12/2021	Sharifi	N/A	G10L 15/22
2022/0189471	12/2021	Sharifi	N/A	G10L 15/22
2023/0061929	12/2022	Carbune	N/A	H04L 63/0861
2023/0088513	12/2022	Karri	704/251	G06N 20/00
2023/0113883	12/2022	Carbune	704/235	G10L 15/28
2024/0013782	12/2023	Siegler	N/A	G10L 15/22

2024/0119088	12/2023	Sharifi	N/A	G06F 3/167
2024/0161741	12/2023	Sharifi	N/A	G06F 3/165
2024/0169995	12/2023	Sharifi	N/A	G10L 17/22
2024/0184989	12/2023	Mann	N/A	G06F 40/166
2024/0304990	12/2023	Tran	N/A	H01Q 3/46
2024/0371372	12/2023	Tasaki	N/A	B60R 16/0373

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3674882	12/2019	EP	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion issued in related PCT Application No. PCT/US2023/036905, dated Feb. 2, 2024. cited by applicant

Primary Examiner: Sonifrank; Richa
Attorney, Agent or Firm: Honigman LLP

Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to multi-user warm words.

BACKGROUND

(2) A user's manner of interacting with an assistant-enabled device is designed primarily, if not exclusively, by means of voice input. For example, a user may ask a device to perform an action including media playback (e.g., music or podcasts), where the device responds by initiating playback of audio that matches the user's criteria. In instances where a device (e.g., a smart speaker) is commonly shared by multiple users in an environment, the device may need to field multiple actions requested by the users that may compete with one another.

SUMMARY

(3) One aspect of the disclosure provides a computer-implemented method that when executed on data processing hardware causes the data processing hardware to perform operations that include detecting a presence of multiple users within an environment of an assistant-enabled device (AED) executing a digital assistant, and for each user of the multiple users, obtaining a respective active set of warm words that each specify a respective action for the digital assistant to perform. Based on the respective active set of warm words for each user of the multiple users, the operations also include executing a warm word arbitration routine to enable a final set of warm words for detection by the AED. The final set of warm words enabled for detection by the AED include warm words selected from the respective active set of warm words for at least one user of the multiple users detected within the environment of the AED. While the final set of warm words are enabled for detection by the AED, the operations also include receiving audio data corresponding to an utterance captured by the AED, detecting, in the audio data, a warm word from the final set of warm words, and instructing the digital assistant to perform the respective action specified by the detected warm word.

(4) Implementations of the disclosure may include one or more of the following optional features. In some implementations, detecting the presence of the multiple users within the environment of the AED includes detecting at least one of the multiple users is present within the environment based on proximity information of the AED relative to a user device associated with the at least one

of the multiple users. In additional implementations, detecting the presence of the multiple users within the environment of the AED includes receiving image data corresponding to a scene of the environment and detecting the presence of at least one of the multiple users within the environment based on the image data. In some examples, detecting the presence of the multiple users within the environment of the AED includes detecting the presence of a corresponding user within the environment of the AED based on receiving voice data characterizing a voice query issued by the corresponding user and directed toward the digital assistant, performing speaker identification on the voice data to identify the corresponding user that issued the voice query, and determining that the identified corresponding user that issued the voice query is present within the environment of the AED. In these examples, the voice query issued by the corresponding user comprises a command that causes the digital assistant to perform a long-standing operation specified by the command and obtaining the respective active set of warm words includes adding, in response to the digital assistant performing the long-standing operation specified by the command, to the respective active set of warm words for the corresponding user that issued the voice query, one or more warm words specifying respective actions for controlling the long-standing operation.

(5) In some implementations, the operations also include detecting the presence of a new user within the environment of the AED, wherein the warm word arbitration routine executes in response to detecting the presence of the new user within the environment. The operations may optionally include determining that one of the multiple users is no longer present within the environment of the AED, wherein the warm word arbitration routine executes in response to determining that the one of the multiple users is no longer present within the environment. In some examples, the operations also include determining addition of a new warm word to, or removal of one of the warm words from, the respective active set of warm words for one of the multiple users present in the environment of the AED, wherein the warm word arbitration routine executes in response to determining the addition of the new warm word to, or the removal of the one of the warm words from, the respective active set of warm words for the one of the multiple users present in the environment of the AED. In some implementations, the operations also include determining a change in ambient context of the AED, wherein the warm word arbitration routine executes in response to determining the change in ambient context.

(6) In some examples, executing the warm word arbitration routine includes obtaining enabled warm word constraints and determining a number of warm words to enable in the final set of warm words for detection by the AED based on the enabled warm word constraints. Here, the enabled warm word constraints include at least one of memory and computing resource availability on the AED for detection of warm words, computational requirements for enabling each warm word in the respective active set of warm words for each user of the multiple users present in the environment of the AED, an acceptable false accept rate tolerance, or an acceptable false reject rate tolerance. In some implementations, executing the warm word arbitration routine includes ranking the warm words in the respective active set of warm words from highest priority to lowest priority based on warm word prioritization signals for each corresponding user of the multiple users, and enabling the final set of warm words for detection by the AED based on the rankings of the warm words in the respective active set of warm words for each corresponding user of the multiple users. In some examples, executing the warm word arbitration routine includes identifying any shared warm words corresponding to warm words present in the respective active set of warm words for each of at least two of the multiple users and determining the final set of warm words is based on assigning, for inclusion in the final set of warm words, a higher priority to warm words identified as shared warm words.

(7) In some implementations, executing the warm word arbitration routine includes determining a warm word affinity score for each user and enabling the final set of warm words for detection by the AED based on the warm word affinity score determined for each corresponding user. Here, the warm word affinity score determined for each corresponding user may be based on at least one of:

frequency of warm word usage by the corresponding user; frequency of interactions between the corresponding user and the digital assistant; duration of presence of the corresponding user in the environment of the AED; a proximity of the user relative to the AED; or a current user context.

(8) In some examples, the operations further include determining that the warm word detected in the audio data corresponding to the utterance captured by the user device comprises a speaker-specific warm word selected from the respective active set of warm words for each of one or more users among the multiple users such that the digital assistant only performs the respective action specified by the speaker-specific warm word when the speaker-specific warm word is spoken by any of the one or more users corresponding to the respective active set of warm words, and based on determining that the warm word detected in the audio data corresponding to the utterance captured by the user device includes a speaker-specific warm word, performing speaker verification on the audio data to determine that the utterance was spoken by one of the one or more users corresponding to the respective active set of warm words that the detected warm word was selected. Here, instructing the digital assistant to perform the respective action specified by the detected warm word is based on the speaker verification performed on the audio data.

(9) In some implementations, final set of warm words are enabled for detection by activating, for each warm word in the final set of warm words, a respective warm word model to run on the assistant-enabled device, and detecting, in the audio data, the warm word from the final set of warm words includes detecting, using the activated respective warm word model, the warm word in the audio data without performing speech recognition on the audio data. In these implementations, detecting the warm word in the audio data may include extracting audio features of the audio data, generating, using the activated respective warm word model, a warm word confidence score by processing the extracted audio features, and determining that the audio data corresponding to the utterance includes the warm word when the warm word confidence score satisfies a warm word confidence threshold. The final set of warm words may be enabled for detection by executing a speech recognizer on the AED, the speech recognizer biased to recognize the warm words in the final set of warm words, and detecting, in the audio data, the warm word from the final set of warm words may include recognizing, using the speech recognizer executing on the AED, the repeat warm word in the audio data.

(10) Another aspect of the disclosure provides a system including data processing hardware and memory hardware in communication with the data processing hardware. The memory hardware stores instructions that when executed by the data processing hardware cause the data processing hardware to perform operations that include detecting a presence of multiple users within an environment of an assistant-enabled device (AED) executing a digital assistant, and for each user of the multiple users, obtaining a respective active set of warm words that each specify a respective action for the digital assistant to perform. Based on the respective active set of warm words for each user of the multiple users, the operations also include executing a warm word arbitration routine to enable a final set of warm words for detection by the AED. The final set of warm words enabled for detection by the AED include warm words selected from the respective active set of warm words for at least one user of the multiple users detected within the environment of the AED. While the final set of warm words are enabled for detection by the AED, the operations also include receiving audio data corresponding to an utterance captured by the AED, detecting, in the audio data, a warm word from the final set of warm words, and instructing the digital assistant to perform the respective action specified by the detected warm word.

(11) This aspect may include one or more of the following optional features. In some implementations, detecting the presence of the multiple users within the environment of the AED includes detecting at least one of the multiple users is present within the environment based on proximity information of the AED relative to a user device associated with the at least one of the multiple users. In additional implementations, detecting the presence of the multiple users within the environment of the AED includes receiving image data corresponding to a scene of the

environment and detecting the presence of at least one of the multiple users within the environment based on the image data. In some examples, detecting the presence of the multiple users within the environment of the AED includes detecting the presence of a corresponding user within the environment of the AED based on receiving voice data characterizing a voice query issued by the corresponding user and directed toward the digital assistant, performing speaker identification on the voice data to identify the corresponding user that issued the voice query, and determining that the identified corresponding user that issued the voice query is present within the environment of the AED. In these examples, the voice query issued by the corresponding user comprises a command that causes the digital assistant to perform a long-standing operation specified by the command and obtaining the respective active set of warm words includes adding, in response to the digital assistant performing the long-standing operation specified by the command, to the respective active set of warm words for the corresponding user that issued the voice query, one or more warm words specifying respective actions for controlling the long-standing operation.

(12) In some implementations, the operations also include detecting the presence of a new user within the environment of the AED, wherein the warm word arbitration routine executes in response to detecting the presence of the new user within the environment. The operations may optionally include determining that one of the multiple users is no longer present within the environment of the AED, wherein the warm word arbitration routine executes in response to determining that the one of the multiple users is no longer present within the environment. In some examples, the operations also include determining addition of a new warm word to, or removal of one of the warm words from, the respective active set of warm words for one of the multiple users present in the environment of the AED, wherein the warm word arbitration routine executes in response to determining the addition of the new warm word to, or the removal of the one of the warm words from, the respective active set of warm words for the one of the multiple users present in the environment of the AED. In some implementations, the operations also include determining a change in ambient context of the AED, wherein the warm word arbitration routine executes in response to determining the change in ambient context.

(13) In some examples, executing the warm word arbitration routine includes obtaining enabled warm word constraints and determining a number of warm words to enable in the final set of warm words for detection by the AED based on the enabled warm word constraints. Here, the enabled warm word constraints include at least one of memory and computing resource availability on the AED for detection of warm words, computational requirements for enabling each warm word in the respective active set of warm words for each user of the multiple users present in the environment of the AED, an acceptable false accept rate tolerance, or an acceptable false reject rate tolerance. In some implementations, executing the warm word arbitration routine includes ranking the warm words in the respective active set of warm words from highest priority to lowest priority based on warm word prioritization signals for each corresponding user of the multiple users, and enabling the final set of warm words for detection by the AED based on the rankings of the warm words in the respective active set of warm words for each corresponding user of the multiple users. In some examples, executing the warm word arbitration routine includes identifying any shared warm words corresponding to warm words present in the respective active set of warm words for each of at least two of the multiple users and determining the final set of warm words is based on assigning, for inclusion in the final set of warm words, a higher priority to warm words identified as shared warm words.

(14) In some implementations, executing the warm word arbitration routine includes determining a warm word affinity score for each user and enabling the final set of warm words for detection by the AED based on the warm word affinity score determined for each corresponding user. Here, the warm word affinity score determined for each corresponding user may be based on at least one of: frequency of warm word usage by the corresponding user; frequency of interactions between the corresponding user and the digital assistant; duration of presence of the corresponding user in the

environment of the AED; a proximity of the user relative to the AED; or a current user context.

(15) In some examples, the operations further include determining that the warm word detected in the audio data corresponding to the utterance captured by the user device comprises a speaker-specific warm word selected from the respective active set of warm words for each of one or more users among the multiple users such that the digital assistant only performs the respective action specified by the speaker-specific warm word when the speaker-specific warm word is spoken by any of the one or more users corresponding to the respective active set of warm words, and based on determining that the warm word detected in the audio data corresponding to the utterance captured by the user device includes a speaker-specific warm word, performing speaker verification on the audio data to determine that the utterance was spoken by one of the one or more users corresponding to the respective active set of warm words that the detected warm word was selected. Here, instructing the digital assistant to perform the respective action specified by the detected warm word is based on the speaker verification performed on the audio data.

(16) In some implementations, final set of warm words are enabled for detection by activating, for each warm word in the final set of warm words, a respective warm word model to run on the assistant-enabled device, and detecting, in the audio data, the warm word from the final set of warm words includes detecting, using the activated respective warm word model, the warm word in the audio data without performing speech recognition on the audio data. In these implementations, detecting the warm word in the audio data may include extracting audio features of the audio data, generating, using the activated respective warm word model, a warm word confidence score by processing the extracted audio features, and determining that the audio data corresponding to the utterance includes the warm word when the warm word confidence score satisfies a warm word confidence threshold. The final set of warm words may be enabled for detection by executing a speech recognizer on the AED, the speech recognizer biased to recognize the warm words in the final set of warm words, and detecting, in the audio data, the warm word from the final set of warm words may include recognizing, using the speech recognizer executing on the AED, the repeat warm word in the audio data.

(17) The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

(1) FIGS. 1A-1C are schematic views of an example system including multiple users having respective active sets of warm words that each specify a respective action for a digital assistant to perform.

(2) FIG. 2 is an example data store storing enrolled user data.

(3) FIGS. 3A and 3B are example graphical user interfaces (GUIs) rendered on a screen of a user device.

(4) FIG. 4 is a schematic view of an example warm word arbitration process for enabling a final set of warm words for detection on an assistant-enabled device.

(5) FIG. 5 is a schematic view of a speaker identification process.

(6) FIG. 6 is a flowchart of an example arrangement of operations for a method of enabling a final set of warm words for detection on an assistant-enabled device when a presence of multiple users is detected in an environment of the assistant-enabled device.

(7) FIG. 7 is a schematic view of an example computing device that may be used to implement the systems and methods described herein.

(8) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

(9) A user's manner of interacting with an assistant-enabled device is designed primarily, if not exclusively, by means of voice input. For example, a user may ask a device to perform an action including media playback (e.g., music or podcasts), where the device responds by initiating playback of audio that matches the user's criteria. Consequently, the assistant-enabled device must have some way of discerning when any given utterance in a surrounding environment is directed toward the device as opposed to being directed to an individual in the environment or originating from a non-human source (e.g., a television or music player). One way to accomplish this is to use a hotword, which by agreement among the users in the environment, is reserved as a predetermined word(s) that is spoken to invoke the attention of the device. In an example environment, the hotword used to invoke the assistant's attention are the words "OK computer." Consequently, each time the words "OK computer" are spoken, it is picked up by a microphone, conveyed to a hotword detector, which performs speech modeling techniques to determine whether the hotword was spoken and, if so, awaits an ensuing command or query. Accordingly, utterances directed at an assistant-enabled device take the general form [HOTWORD] [QUERY], where "HOTWORD" in this example is "OK computer" and "QUERY" can be any question, command, declaration, or other request that can be speech recognized, parsed and acted on by the system, either alone or in conjunction with the server via the network.

(10) In cases where the user provides a sequence of several hotword based commands to an assistant-enabled device, such as a mobile phone or smart speaker, the user's interaction with the phone or speaker may become awkward. The user may speak, "Ok computer, play my homework playlist." The phone or speaker may begin to play the first song on the playlist. The user may wish to advance to the next song and speak, "Ok computer, next." To advance to yet another song, the user may speak, "Ok computer, next," again. To alleviate the need to keep repeating the hotword before speaking a command, the assistant-enabled device may be configured to recognize/detect a narrow set of hotphrases or warm words to directly trigger respective actions. In the example, the warm word "next" serves the dual purpose of a hotword and a command so that the user can simply utter "next" to invoke the assistant-enabled device to trigger performance of the respective action instead of uttering "Ok computer, next." Other non-limiting warm words and hotphrases may include "what's the weather?", "set a timer", "volume up", and "volume down".

(11) A set of warm words can be active for controlling a long-standing operation. As used herein, a long-standing operation refers to an application or event that a digital assistant performs for an extended duration and one that can be controlled by the user while the application or event is in progress. For instance, when a digital assistant sets a timer for 30-minutes, the timer is a long-standing operation from the time of setting the timer until the timer ends or a resulting alert is acknowledged after the timer ends. In this instance, a warm word such as "stop timer" could be active to allow the user to stop the timer by simply speaking "stop timer" without first speaking the hotword. Likewise, a command instructing the digital assistant to play music from a streaming music service is a long-standing operation while the digital assistant is streaming music from the streaming music service through a playback device. In this instance, an active set of warm words can be "pause", "pause music", "volume up", "volume down", "next", "previous", etc., for controlling playback of the music the digital assistant is streaming through the playback device. The long-standing operation may include multi-step dialogue queries such as "book a restaurant" in which different sets of warm words will be active depending on a given stage of the multi-step dialogue. For instance, the digital assistant may prompt a user to select from a list of restaurants, whereby a set of warm words may become active that each include a respective identifier (e.g., restaurant name or number in list) for selecting a restaurant from the list and complete the action of booking a reservation for that restaurant.

(12) One challenge with warm words is limiting the number of words/phrases that are simultaneously active so that quality and efficiency is not degraded. For instance, a number of false

positives indicating when the assistant-enabled device incorrectly detected/recognized one of the active words greatly increases the larger the number of warm words that are simultaneously active. Moreover a user that seeds a command to initiate a long-running operation cannot prevent others from speaking active warm words for controlling the long-running operation.

(13) Since warm words and/or hotphrases may be active for extended periods of time (e.g., always-on), there may often be a limit in terms of how many different words/phrases may be enabled for detection by an assistant-enabled device (AED) at any given time. For instance, a computational budget based on computing resource constraints due to processing capabilities and memory availability on the AED may impact how many different warm words can be enabled at any given time. Considerations of computational budget is especially important for battery-powered devices since the models for recognizing/detecting warm words and/or hotphrases typically run on digital signal processors (DSPs). Another challenge with warm words is limiting the number of words/phrases that are simultaneously enabled so that quality and efficiency is not degraded. For instance, a number of false positives (i.e., also referred to as ‘false alarm rate’) indicating when the AED incorrectly detected/recognized one of the warm words greatly increases the larger the number of warm words that are simultaneously enabled for detection by the AED.

(14) As the total number of different warm words that can be enabled for detection is typically limited due to the one or more of the factors mentioned above, difficulties on shared AEDs that aim to serve multiple users (e.g., household family members sharing an assistant-enabled device) arise since different users may have divergent preferences that each may depend on different sets of warm words being active. For instance, one user may wish to issue commands for controlling music playback on the AED, while another user sharing the same AED may wish to issue messaging commands for facilitating communication of messages between that user and some remote recipient. Moreover, some of the users may have different tolerance preferences with respect to consequences of false-accepting (and/or false-rejecting) detection of warm words. For instance, some users may tolerate an over-trigger of “stop” while other users may not. Ideally, the AED aims to enable as many of the warm words in each active set of warm words based on the computational budget of the AED at any given time.

(15) Implementations herein are directed toward detecting a presence of multiple users within an environment of the AED and obtaining the respective active set of warm words that each specify a respective action for the digital assistant to perform. Based on the respective active set of warm words obtained for each user, implementations are further directed toward executing a warm word arbitration routine to enable a final set of warm words for detection by the AED whereby the final set of warm words include warm words selected from the respective active set of warm words for at least one of user among the multiple users detected within the environment of the AED

(16) As used herein, an “active” warm word includes a warm word that a user prefers to have the AED enable for detection/recognition without requiring the user to speak a predefined hotword to “wake-up” the AED, whereas a warm word in the final set of warm words enabled for detection is affirmatively enabled for the AED to detect/recognize. Thus, as long as the computational budget of the AED permits, the warm word arbitration routine will aim to include all of the active warm words in the final set of warm words enabled for detection by the AED. Otherwise, when a total number of warm words permissible in the final set of warm words is less than a total number of active warm words from the respective active sets of warm words, the warm word arbitration routine is tasked with selecting those active warm words for inclusion in the final set of warm words that are ranked with higher priorities than those not selected for inclusion in the final set of warm words. As a result, there may be conditions and scenarios where a warm word found in an active set of warm words for a given user may ultimately not be included in the final set of warm words, and thus, not enabled for detection by the AED such that unselected warm word would not be recognized/detected when spoken in an utterance unless the utterance also included a predefined hotword (e.g., “Hey Computer). Accordingly, the warm word arbitration routine aims to

dynamically select warm words for inclusion in the final set of warm words on an ongoing basis in order to maximize the total number of warm words permissible in the final set of warm words enabled for detection by the AED.

(17) FIGS. 1A-1C illustrate an example system **100** for enabling a final set of warm words **112F** for detection by an assistant-enabled device (AED) **104**, wherein the final set of warm words **112F** include warm words **112** selected from a respective active set of warm words **112A** for at least one user among multiple users detected within an environment of the AED **104**. The AED **104** may execute one or more digital assistants **105** and each warm word **112** (whether present in one of the active sets and/or the final set of warm words **112A**, **112F**) specifies a respective action for at least one of the one or more digital assistants **105** to perform. The users **102** may interact with a digital assistant **105** through speech. For simplicity, examples herein depict the AED **104** executing a single digital assistant **105**. However, the present disclosure is not limited to the number of digital assistants **105** the AED **104** is capable of executing at any given time such that the AED **104** can execute any combination of digital assistants concurrently or individually at any given time.

(18) In some implementations, for each warm word **112** in the final set of warm words **112F**, the AED **104** additionally receives a respective warm word model **330** configured to detect the corresponding warm word **112** in streaming audio without performing speech recognition. For example, the AED **104** (and/or the server **130**) additionally includes one or more warm word models **330**. Here, the warm word models **330** may be stored on the memory hardware **12** of the AED **104** or the remote memory hardware **134** on the server **130**. If stored on the server **130**, the AED **104** may request the server **130** to retrieve a warm word model **330** for a corresponding warm word **112** and provide the retrieved warm word model **330** so that the AED **104** (via the warm word arbitration routine **401**) can enable the warm word model **330**. An enabled warm word model **330** running on the AED **104** may detect an utterance of the corresponding warm word **112** in streaming audio captured by the AED **104** without performing speech recognition on the captured audio. Further, a single warm word model **330** may be capable of detecting all of the warm words **112** from the final set of warm words **112F** in streaming audio.

(19) In some configurations, the AED **104** receives code associated with an application loaded on the AED **104** (e.g., a music application running in the foreground or background of the AED **104**) to identify any warm words **112** and associated warm word models **330** that developers of the application want users **102** to be able to speak to interact with the application and the respective actions for each warm word **112**. In other examples, the AED **104** receives, for at least one warm word **112** in the respective active set of warm words **112A** for at least one user **102**, via a warm word application programming interface (API) executing on the AED **104**, a respective warm word model **330** configured to detect the corresponding warm word **112** in streaming audio without performing speech recognition. The warm words **112** in the registry may also relate to follow-up queries that the user **102** (or typical users) tend to issue following the given query, e.g., “Ok computer, play my music playlist”.

(20) In additional implementations, enabling the final set of warm words **112F** causes the AED **104** to execute the speech recognizer **116** in a low-power and low-fidelity state. Here, the speech recognizer **116** is constrained or biased to only recognize the warm words in the final set of warm words **112F** when spoken in utterances captured by the AED **104**. Since the speech recognizer **116** is only recognizing a limited number of terms/phrases, the number of parameters of the speech recognizer **116** may be drastically reduced, thereby reducing the memory requirements and number of computations needed for recognizing the active warm words in speech. Accordingly, the low-power and low-fidelity characteristics of the speech recognizer **116** may be suitable for execution on a digital signal processor (DSP). In these implementations, the speech recognizer **116** executing on the AED **104** may recognize an utterance **106** of a warm word **112** in streaming audio captured by the AED **104** in lieu of using a warm word model **330**. In some examples, detection of a warm word **112** by a corresponding warm word model **330** is confirmed by the speech recognizer **116**

performing speech recognition on the audio data.

(21) In the example shown, the AED **104** includes a smart speaker. However, the AED **104** can include other computing devices, such as, without limitation, a smart phone, tablet, smart display, headset, desktop/laptop, smart watch, smart appliance, headphones, other wearables, or vehicle infotainment device. The AED **104** includes data processing hardware **10** and memory hardware **12** storing instructions that when executed on the data processing hardware **10** cause the data processing hardware **10** to perform operations. The AED **104** includes an array of one or more microphones **16** configured to capture acoustic sounds such as speech directed toward the AED **104**. The AED **104** may also include, or be in communication with, an audio output device (e.g., speaker) **18** that may output audio such as music **122** and/or synthesized speech from the digital assistant **105**. In some configurations, the AED **104** also includes, or is in communication with, a display **13** configured to display content from various sources. Moreover, the AED **104** may include, or be in communication with, one or more cameras **19** configured to capture images within the environment and output image data **412** (FIG. 4).

(22) In some configurations, the AED **104** is in communication with multiple user devices **50** associated with the multiple users **102**. In the examples shown, the second and third users **102b**, **102c** each include a respective user device **50** that includes a smart phone that the respective user **102** may interact with. However, the user device **50** can include other computing devices, such as, without limitation, a smart watch, smart display, smart glasses, a smart phone, smart glasses/headset, tablet, smart appliance, headphones, a computing device, a smart speaker, or another assistant-enabled device. Each user device **50** may include at least one microphone **52** residing on the user device **50** and in communication with the AED **104**. In these configurations, the user device **50** may also be in communication with the one or more microphones **16** residing on the AED **104**. Additionally, the multiple users **102** may control and/or configure the AED **104**, as well as interact with the digital assistant **105**, using an interface **200**, such as a graphical user interface (GUI) **200** rendered for display on a respective screen of each user device **50**.

(23) With continued reference to FIGS. 1A-1C and 4, during execution of the digital assistant **105**, the AED **104** detects the multiple users **102a-c** in the environment using a user detector **410** and obtains a respective active set of warm words **112A** for each user **102** detected in the environment. Based on the respective active set of warm words **112A** for each of the multiple users **102** detected in the environment of the AED **104**, a warm word selector **400** running on the AED **104** selects a final set of warm words **112F** to enable for simultaneous detection on the AED **104**. For example, the warm word selector **400** receives proximity information **54** (FIG. 4) for the location of each of the multiple users **102a-c** relative to the AED **104** via the user detector **410**. In some implementations, for one or more of the users **102** having a respective user device **50**, the respective user device **50** is capable of broadcasting proximity information **54** receivable by the user detector **410** that the AED **104** uses to determine the proximity of each user device **50** relative to the AED **104**. The proximity information **54** from each user device **50** may include wireless communication signals, such as WiFi, Bluetooth, or Ultrasonic, in which the signal strength of the wireless communication signals received by the user detector **410** may correlate proximities (e.g. distances) of the user device **50** relative to the AED **104**. The proximity information **54** received from each user device **50** may include a device identifier **50** that uniquely identifies the device **50** and that may be used by the user detector **410** to resolve an identity of the user **102** using conventional techniques.

(24) In additional implementations, the user detector **410** automatically detects one or more of the multiple users **102** in the environment by receiving image data **412** corresponding to a scene of the environment and obtained by the camera **19**. Here, the user detector **410** detects the multiple users **102** based on the received image data **312**. The user detector **410** may detect the users **102** based on the image data **412** anonymously without uniquely identifying the users **102**. In some implementations, the user detector **410** performs facial recognition on the received image data **312**

and attempts to uniquely identify each user **102** based on the facial recognition performed. In these implementations, each user **102** expressly grants privileges to the digital assistant **105** to perform facial recognition and each user **102** has the option to revoke the granted privilege at any time.

(25) Similarly, the user detector **410** may detect the multiple users **102** in the environment by performing speaker identification (FIG. 5) to resolve the identity of a user **102** within the environment. Here, the user detector **410** may detect one or more of the multiple users **102** based on received audio data **502** associated with the issued queries, and the user detector **410** may continue to detect a user **102** that issued a query (and associated audio data **502**) for a threshold amount of time after the user **102** spoke. Notably, the user detector **510** may use any combination of techniques to obtain results that may be correlated to detect the number of different users **102** within the environment of the AED **102**. Each user device **50** may broadcast a current active set of warm words **112A** for the associated user **102** that the warm word selector (and warm word arbitration routine **401**) receives. Similarly, the active set of warm words **112A** for one or more of the detected users **102** may be retrieved from profile information associated with a detected user once an identity of the user **102** is resolved.

(26) In some implementations, the user detector **410** resolves an identity of each of the multiple users. In some scenarios, a user **102** is identified as an enrolled user **200** of the AED **104** that is authorized to access or control various functions of the AED **104** and digital assistant **105**. The AED **104** may have multiple different enrolled users **200** each having registered user accounts indicating particular permissions or rights regarding functionality of the AED **104**. For instance, the AED **104** may operate in a multi-user environment such as a household with multiple family members, whereby each family member corresponds to an enrolled user **200** having permissions for accessing a different respective set of resources. To illustrate, a mother named Barb speaking the command “play my music playlist” would result in the digital assistant **105** streaming music from a rock music playlist associated with the mother, as opposed to a different music playlists created by, and associated with, another enrolled user **200** of the household such as a teenage daughter whose playlist includes pop music.

(27) FIG. 2 shows an example data store storing enrolled user data/information for each of multiple enrolled users **200a-n** of the AED **104**. Here, each enrolled user **200** of the AED **104** may undertake a voice enrollment process to obtain a respective enrolled speaker vector **154** from audio samples of multiple enrollment phrases spoken by the enrolled user **200**. For example, a speaker-discriminative model **510** (FIG. 5) may generate one or more enrolled speaker vectors **154** from the audio samples of enrollment phrases spoken by each enrolled user **200** that may be combined, e.g., averaged or otherwise accumulated, to form the respective enrolled speaker vector **154**. One or more of the enrolled users **200** may use the AED **104** to conduct the voice enrollment process, where the microphone **16** captures the audio samples of these users speaking the enrollment utterances and the speaker-discriminative model **510** generates the respective enrolled speaker vectors **154** therefrom. The model **510** may execute on the AED **104**, the server **120**, or a combination thereof. Additionally, one or more of the enrolled users **200** may enroll with the AED **104** by providing authorization and authentication credentials to an existing user account with the AED **104**. Here, the existing user account may store enrolled speaker vectors **154** obtained from a previous voice enrollment process with another device also linked to the user account.

(28) In some examples, the enrolled speaker vector **154** for an enrolled user **200** includes a text-dependent enrolled speaker vector. For instance, the text-dependent enrolled speaker vector may be extracted from one or more audio samples of the respective enrolled user **200** speaking a predetermined term such as the hotword **110** (e.g., “Ok computer”) used for invoking the AED **104** to wake-up from a sleep state. In other examples, the enrolled speaker vector **154** for an enrolled user **200** is text-independent obtained from one or more audio samples of the respective enrolled user **200** speaking phrases with different terms/words and of different lengths. In these examples, the text-independent enrolled speaker vector may be obtained over time from audio samples

obtained from speech interactions the user **102** has with the AED **104** or other device linked to the same account.

(29) Additionally, the AED **104** (and/or server **120**) may optionally store one or more other text-dependent speaker vectors **158** each extracted from one or more audio samples of the respective enrolled user **200** speaking a specific term or phrase. For example, the enrolled user **200a** may include a respective text-dependent speaker vector **158** for each of one or more warm words **112** that, when enabled for detection by the AED **104**, may be spoken to cause the AED **104** to perform a respective action for controlling a long-standing operation or perform some other command. Accordingly, a text-dependent speaker vector **158** for a respective enrolled user **200** represents speech characteristics of the respective enrolled user **200** speaking the specific warm word **112**. A text-dependent speaker vector **154** stored for a respective enrolled user **200** that is associated with a specific warm word **112** may be used to verify the respective enrolled user **200** speaking the specific warm word **112** to command the AED **104** to perform an action for controlling a long-standing operation.

(30) FIG. 2 also shows the AED **104** (and/or server **120**) storing warm word preferences **212** for the respective enrolled user **200**. The warm word preferences **212** may include a list of warm words the user **102** selects for inclusion in a respective active set of warm words **112A**. Some of the warm words in the respective active set of warm words **112A** may be activity-based such that the warm words are only “active” depending on contextual information indicating a current activity associated with the user. For instance, activity-based warm words could include music playback settings (e.g., stop, pause, volume up, volume down, etc.) that are only included in the active set of warm words while the AED **104** is streaming music for playback. Here, the contextual information would indicate the current activity of performing the long-standing operation of streaming music for playback from the AED, and thus, would cause the activity-based warm words each specifying a respective action for the digital assistant **105** to control the long-standing operation of music playback from the AED **104** to be activated. In another example, activity-based warm words could include warm words each specifying a respective action for the digital assistant **105** to perform based on context related to a current application **107** the user **102** is currently interacting with. For instance, the user **102** may be interacting with a cooking application **107** executing on a user device **50** associated with the user **102** and the activity-based warm words may relate to actions that need to be performed for a recipe (e.g., preheat oven, set timer) conveyed by the cooking application and/or actions for controlling the cooking application (e.g., next screen) **107** by speech so that the user can navigate the cooking application hands free manner. The list of warm words in the warm word preferences may include preferred warm words that the user **102** selects for inclusion in the respective active set of warm words **112A** independent of any activity the user **102** is undertaking. Here, the preferred warm words are those included in the active set of warm words **112A** that the user **102** desires to speak without speaking a predefined hotword to cause the AED **102** to perform a respective action specified by the warm word. Some preferred warm words could be time-sensitive such that the user **102** defines periods/times/days where the warm words should be active and included in the respective active set of warm words so that the user **102**. For instance, between the times of 9-10 pm when the user **102** is getting ready for bed, the user **102** may wish to speak the warm word “set alarm” to allow the user **102** to set his/her alarm for the next morning without having to first speak a predefined hotword (e.g., Hey Computer).

(31) The warm word preferences **212** stored for each respective enrolled user may further include acceptable false accept rate tolerances and/or acceptable false reject rate tolerances for his/her active set of warm words. These tolerances can dictate how sensitive resulting warm word models are for detecting the presence of the warm words in speech. These tolerances may be included within the enabled warm word constraints **332** received by the warm word arbitration routine **401** when enabling the final set of warm words **112F** for detection by the AED **104**.

(32) FIG. 1A shows the user **102** speaking a first utterance **106**, **106a**, “Ok computer, play my

music playlist” in the vicinity of the AED **104**. The microphone **16** of the AED **104** receives the utterance **106** and processes audio data **502** (FIGS. **4** and **5**) that corresponds to the utterance **106a**. The initial processing of the audio data **402** may involve filtering the audio data **402** and converting the audio data **502** from an analog signal to a digital signal. As the AED **104** processes the audio data **502**, the AED may store the audio data **502** in a buffer of the memory hardware **12** for additional processing. With the audio data **502** in the buffer, the AED **104** may use a hotword detector **108** to detect whether the audio data **402** includes the hotword. The hotword detector **108** is configured to identify hotwords that are included in the audio data **502** without performing speech recognition on the audio data **502**.

(33) In the hotword detector **108** is configured to identify hotwords that are in the initial portion of the utterance **106**. In this example, the hotword detector **108** may determine that the utterance **106** “Ok computer, play my music playlist” includes the hotword **110** “ok computer” if the hotword detector **108** detects acoustic features in the audio data **402** that are characteristic of the hotword **110**. The acoustic features may be mel-frequency cepstral coefficients (MFCCs) that are representations of short-term power spectrums of the utterance **106** or may be mel-scale filterbank energies for the utterance **106**. For example, the hotword detector **108** may detect that the utterance **106** “Ok computer, play music” includes the hotword **110** “ok computer” based on generating MFCCs from the audio data **402** and classifying that the MFCCs include MFCCs that are similar to MFCCs that are characteristic of the hotword “ok computer” as stored in a hotword model of the hotword detector **108**. As another example, the hotword detector **108** may detect that the utterance **106** “Ok computer, play music” includes the hotword **110** “ok computer” based on generating mel-scale filterbank energies from the audio data **402** and classifying that the mel-scale filterbank energies include mel-scale filterbank energies that are similar to mel-scale filterbank energies that are characteristic of the hotword “ok computer” as stored in the hotword model of the hotword detector **108**.

(34) When the hotword detector **108** determines that the audio data **402** that corresponds to the utterance **106** includes the hotword **110**, the AED **104** may trigger a wake-up process to initiate speech recognition on the audio data **402** that corresponds to the utterance **106**. For example, a speech recognizer **116** running on the AED **104** may perform speech recognition or semantic interpretation on the audio data **402** that corresponds to the utterance **106**. The speech recognizer **116** may perform speech recognition on the portion of the audio data **402** that follows the hotword **110**. In this example, the speech recognizer **116** may identify the words “play my music playlist” as a command **118** following the hotword **110**.

(35) In some implementations, the speech recognizer **116** is located on a server **120** in addition to, or in lieu, of the AED **104**. Upon the hotword detector **108** triggering the AED **104** to wake-up responsive to detecting the hotword **110** in the utterance **106**, the AED **104** may transmit the audio data **402** corresponding to the utterance **106** to the server **120** via a network **132**. The AED **104** may transmit the portion of the audio data **402** that includes the hotword **110** for the server **120** to confirm the presence of the hotword **110**. Alternatively, the AED **104** may transmit only the portion of the audio data **402** that corresponds to the portion of the utterance **106** after the hotword **110** to the server **120**. The server **120** executes the speech recognizer **116** to perform speech recognition and returns a transcription of the audio data **402** to the AED **104**. In turn, the AED **104** identifies the words in the utterance **106**, and the AED **104** performs semantic interpretation and identifies any speech commands. The AED **104** (and/or the server **120**) may identify the command for the digital assistant **105** to perform the long-standing operation of “play music”. In the example shown, the digital assistant **105** begins to perform the long-standing operation of playing music **122** as playback audio from the speaker **18** of the AED **104**. The digital assistant **105** may stream the music **122** from a streaming service (not shown) or the digital assistant **105** may instruct the AED **104** to play music stored on the AED **104**.

(36) The AED **104** (and/or the server **120**) may include an operation identifier **124** configured to

identify one or more long-standing operations the digital assistant **105** is currently performing. For each long-standing operation the digital assistant **105** is currently performing, the warm word selector **400** via the warm word arbitration routine **401** may select, for inclusion in the final set of warm words **112F**, a corresponding set of one or more warm words **112** each associated with a respective action for controlling the long-standing operation. In some examples, the warm word selector **400** accesses the warm word preferences **212** from the enrolled user data/information for enrolled users **200** of the AED **104** (e.g., stored on the memory hardware **12**) or another registry or table that associates the identified long-standing operation with a corresponding set of one or more warm words **112** that are highly correlated with the long-standing operation. For example, if the long-standing operation corresponds to a set timer function, the associated set of one or more warm words **112** available the warm word selector **126** to activate includes the warm word **112** “stop timer” for instructing the digital assistant **105** to stop the timer. Similarly, for the long-standing operation of “Call [contact name]” the associated set of warm words **112** includes a “hang up” and/or “end call” warm word(s) **112** for ending the call in progress. In the example shown, for the long-standing operation of playing music **122**, the associated set of one or more warm words **112** available for the warm word selector **126** to activate includes the warm words **112** “next”, “pause”, “previous”, “volume up”, and “volume down” each associated with the respective action for controlling playback of the music **122** from the speaker **18** of the AED **104**. Accordingly, the warm word selector **400** may determine to include these warm words **112** in the final set of warm words **112F** enabled for detection by the AED **104** while the digital assistant **105** is performing the long-standing operation and may disable these warm words **112** once the long-standing operation ends. Similarly, the warm word arbitration routine **401** may enable/disable different warm words **112** depending on a state of the long-standing operation in progress. For example, if the user speaks “pause” to pause the playback of music **122**, the warm word arbitration routine **401** may add a warm word **112** for “play” into the final set of warm words **112F** to resume the playback of the music **122**. In some configurations, instead of accessing a registry or the warm word preferences **212** from the enrolled user data/information for enrolled users **200**, the warm word selector **400** examines code associated with an application of the long-standing operation (e.g., a music application running in the foreground or background of the AED **104**) to identify any warm words **112** that developers of the application want users **102** to be able to speak to interact with the application and the respective actions for each warm word **112**.

(37) In some implementations, after adding the warm words **112** correlated with the long-standing operation for inclusion in the final set of warm words **112F** enabled for detection, the digital assistant **105** associates these warm words **112** with only the user **102** that spoke the utterance **106** with the command **118** for the digital assistant **105** to perform the long-standing operation. That is, the digital assistant **105** configures a portion of the warm words **112** in the final set of warm words **112F** to be speaker-specific such that they are dependent on a speaking voice of the particular user **102** that provided the initial command **118** to initiate the long-standing operation. As will become apparent, by depending warm words **112** on the speaking voice of the particular user **102a**, the AED **104** (e.g., via the digital assistant **105**) will only perform the respective action specified by the one of the warm words **112** when the warm word is spoken by the particular user **102**, and thereby suppress performance (or at least require approval from the particular user **102**) of the respective action when the warm word **112** is spoken by a different speaker.

(38) Referring to FIG. 5, in some examples, the user detector **410** resolves the identity of the user **102** that spoke the utterance **106** by performing a speaker identification process **500**. The speaker identification process **500** may execute on the data processing hardware **12** of the AED **104**. The process **500** may also execute on the server **120**. The speaker identification process **500** identifies the user **102** that spoke the utterance **106** by first extracting, from the audio data **502** corresponding to the utterance **106** spoken by the user **102**, a first speaker-discriminative vector **511** representing characteristics of the utterance **106**. Here, the speaker identification process **500** may execute a

speaker-discriminative model **510** configured to receive the audio data **502** as input and generate, as output, the first speaker-discriminative vector **511**. The speaker-discriminative model **510** may be a neural network model trained under machine or human supervision to output speaker-discriminative vectors **511**. The speaker-discriminative vector **511** output by the speaker-discriminative model **510** may include an N-dimensional vector having a value that corresponds to speech features of the utterance **106** that are associated with the user **102**. In some examples, the speaker-discriminative vector **511** is a d-vector.

(39) Once the first speaker-discriminative vector **511** is output from the model **510**, the speaker identification process **500** determines whether the extracted speaker-discriminative vector **511** matches any of the enrolled speaker vectors **154** stored on the AED **104** (e.g., in the memory hardware **12**) for enrolled users **200a-n** (FIG. 2) of the AED **104**. As described above with reference to FIG. 2, the speaker-discriminative model **510** may generate the enrolled speaker vectors **154** for the enrolled users **200** during a voice enrollment process. Each enrolled speaker vector **154** may be used as a reference vector corresponding to a voiceprint or unique identifier representing characteristics of the voice of the respective enrolled user **200**.

(40) In some implementations, the speaker identification process **500** uses a comparator **520** that compares the first speaker-discriminative vector **511** to the respective enrolled speaker vector **154** associated with each enrolled user **200a-n** of the AED **104**. Here, the comparator **520** may generate a score for each comparison indicating a likelihood that the utterance **106** corresponds to an identity of the respective enrolled user **200**, and the identity is accepted when the score satisfies a threshold. When the score does not satisfy the threshold, the comparator may reject the identity. In some implementations, the comparator **520** computes a respective cosine distance between the first speaker-discriminative vector **511** and each enrolled speaker vector **154** and determines the first speaker-discriminative vector **511** matches one of the enrolled speaker vectors **154** when the respective cosine distance satisfies a cosine distance threshold.

(41) In some examples, the first speaker-discriminative vector **511** is a text-dependent speaker-discriminative vector extracted from a portion of the audio data that includes the hotword **110** and each enrolled speaker vector **154** is also text-dependent on the same hotword **110**. The use of text-dependent speaker vectors can improve accuracy in determining whether the first speaker-discriminative vector **511** matches any of the enrolled speaker vectors **154**. In other examples, the first speaker-discriminative vector **511** is a text-independent speaker-discriminative vector extracted from the entire audio data that includes both the hotword **110** and the command **118** or from the portion of the audio data that includes the command **118**.

(42) When the speaker identification process **500** determines that the first speaker-discriminative vector **511** matches one of the enrolled speaker vectors **154**, the process **500** identifies the user **102** that spoke the utterance **106** as the respective enrolled user **200** associated with the one of the enrolled speaker vectors **154** that matches the extracted speaker-discriminative vector **511**. In the example shown, the comparator **520** determines the match based on the respective cosine distance between the first speaker-discriminative vector **511** and the enrolled speaker vector **154** associated with the first enrolled user **200a** satisfying a cosine distance threshold. In some scenarios, the comparator **520** identifies the user **102** as the respective first enrolled user **200a** associated with the enrolled speaker vector **154** having the shortest respective cosine distance from the first speaker-discriminative vector **511**, provided this shortest respective cosine distance also satisfies the cosine distance threshold.

(43) Conversely, when the speaker identification process **500** determines that the first speaker-discriminative vector **511** does not match any of the enrolled speaker vectors **154**, the process **500** may identify the user **102** that spoke the utterance **106** as a guest user of the AED **104**. Accordingly, the user detector **410** may associate the activated set of one or more warm words **112** with the guest user and use the first speaker-discriminative vector **511** as a reference speaker vector representing the speech characteristics of the voice of the guest user. In some instances, the guest user could

enroll with the AED **104** and the AED **104** could store the first speaker-discriminative vector **511** as a respective enrolled speaker vector **154** for the newly enrolled user.

(44) In the example shown in FIG. **1A**, the AED **104** notifies the identified user **102** (e.g., Barb) associated with the respective active set of warm words **112A** for controlling the long-standing operation that the warm words **112** are enabled and that the user **102** can speak any of the warm words **112** to instruct the AED **104** to perform the respective action for controlling the long-standing operation. For instance, the digital assistant **105** may generate synthesized speech **123** for audible output from the speaker **18** of the AED **104** that states, “Barb, you may speak music playback controls without saying ‘Ok computer’”. In additional examples, the digital assistant **105** may provide a notification to a user device **50** (e.g., smart phone) linked to the user account of the identified user to inform the identified user **102** (e.g., Barb) which warm words **112** are currently active for controlling the long-standing operation.

(45) Referring to FIG. **3A**, a graphical user interface (GUI) **300**, **300a** executing on the user device **50** may display the enabled warm words **112** and associated respective actions for controlling the long-standing operation. Notably, the GUI **300a** of FIG. **3A** depicts the presentation of a screen related to controlling a long-standing operation initiated by a user **102a**, and therefore, presents only the warm words and associated actions related to control of the long-standing operation in progress. Thus, there may be additional warm words in the respective activated set of warm words **112A** for the user **102**. For instance, FIG. **1A** also shows the warm words **112** ‘Call [Contact]’, ‘Turn off lights’, and ‘Turn on lights’ included in the respective active set of warm words **112A** for the user **102a**, albeit not related to the long-standing operation of playback of Barb's music playlist. Each warm word itself may serve as a descriptor identifying the respective action. FIG. **3A** provides an example GUI **300a** displayed on a screen of the user device **50** to inform the user **102** which warm words **112** are active for that user **102** to speak to control the long-standing operation, and which warm words **112N** are not enabled or simply inactive/disabled, and thus, unavailable for controlling the long-standing operation when spoken by the user **102**. Specifically, the GUI **300a** renders the active warm words **112** “next”, “pause”, “previous”, “volume up”, and “volume down,” and the not enabled warm word **112N** “play”. If the user **102** were to pause playback of the music, the warm word for “play” may become an active warm word **112** and the warm word for “pause” may become an inactive warm word **112N**. Each warm word **112** is associated with the respective action for controlling playback of the music **122** from the speaker **18** of the AED **104**.

(46) Additionally, the GUI **300a** may render for display an identifier of the long-standing operation (e.g., “Playing Track **1**”), an identifier of the AED **104** (e.g., smart speaker) that is currently performing the long-standing operation, and/or an identity of the active user **102** (e.g., Barb) that initiated the long-standing operation. In some implementations, the identity of the active user **102** includes an image **304** of the active user **102**. Accordingly, by identifying the active user **102** and the active warm words **112**, the GUI **300a** reveals the active user **102** as a “controller” of the long-standing operation that may speak any of the active warm words **112** displayed in GUI **300a** to perform a respective action for controlling the long-standing operation. As mentioned above, the warm words in the active set of warm words **112** related to controlling the long-standing operation may optionally be dependent on the speaking voice of Barb **102**, since Barb **102** seeded the initial command **118** “play music,” to initiate the long-standing operation. By depending the active set of warm words **112** on the speaking voice of Barb **102**, the AED **104** (e.g., via the digital assistant **105**) will only perform a respective action associated with one of the warm words **112** when the warm word **112** is spoken by Barb **102**, and will suppress performance (or at least require approval from Barb **102**) of the respective action when the warm word **112** is spoken by a different speaker.

(47) The user device **50** may also render graphical elements **302** for display in the GUI **300a** for performing the respective actions associated with the respective active warm words **112** to playback of the music **122** from the speaker **18** of the AED **104**. In the example shown, the graphical elements **302** are associated with playback controls for the long-standing operation of

playing music **122**, that when selected, cause the device **50** to perform a respective action. For instance, the graphical elements **302** may include playback controls for performing the action associated with the warm word **112** “next,” performing the action associated with the warm word **112** “pause,” performing the action associated with the warm word **112** “previous,” performing the action associated with warm word **112** “volume up,” and performing the action associated with the warm word **112** “volume down.” The GUI **300a** may receive user input indications via any one of touch, speech, gesture, gaze, and/or an input device (e.g., mouse or stylist) to control the playback of the music **122** from the speaker **18** of the AED **104**. For example the user **102** may provide a user input indication indicating selection of the “next” control (e.g., by touching the graphical button in the GUI **300a** that universally represents “next”) to cause the AED **104** to perform the action of advancing to the next song in the playlist associated with the music **122**.

(48) Referring to FIG. 3B, a GUI **300**, **300b** executing on the user device **50** (or the AED **104**) of the second user **102b** (e.g., Jim) displays a warm word configurator screen that allows the user **102b** to select which warm words **112** to add to the user's **102b** respective active set of warm words **112A**, **112Ab** (FIGS. 1A-1C). In the example shown, the GUI **300b** presents a list of available warm words **112** displayed as respective graphical elements that may be selected to add or remove the corresponding available warm word to the respective active set of warm words **112A** that the user wants the AED **104** to detect, and ultimately perform the respective action specified by the corresponding warm word, when spoken by the user **102b**. The GUI **300b** provides the ability to select activity-based warm words that the user **102b** wants to activate by adding to the respective active set of warm words **112A** when the user **102b** is undertaking a particular activity. Here, the particular activity may be ascertained based on a current application executing on the user device **50** or AED **104** that the user **102b** is interacting with. For instance, the warm word configurator screen presented by the GUI **300b** may provide a list of applications loaded on the user device **50** and list, for each corresponding application, available activity-based warm words that the user **102** may select to add to the respective active set of warm words **112A** when the user **102** is interacting with the corresponding application. A non-exhaustive list of applications each having available activity-based warm words may include a cooking application and a music player application. In the example shown, the user may select the activity-based warm words ‘pre-heat oven’, and ‘set timer’ that may be added to the respective active set of warm words **112Ab** for the user **102b** when the user **102** is interacting with the cooking application. For instance, in FIG. 1A the user may be viewing a recipe in the cooking app for baking deboned chicken, and when the user is viewing recipe preparation steps the activity-based warm words ‘pre-heat oven’ may be added to the respective active set of warm words **112Ab**. Here, the digital assistant **105** instructs a smart oven to turn-on and pre-heat when the user **102b** when the AED **104** detects the warm word ‘pre-heat oven’ in audio data characterizing an utterance spoken by the user **102b** and captured by the AED **104**. Notably, and as described in greater detail below, the warm word arbitration routine **401** must affirmatively choose to enable the warm word ‘pre-heat oven’ as one of the final set of warm words **112F** in order to permit the AED **104** to detect the warm word in audio data and perform the respective action.

(49) With continued reference to the GUI **300b** of FIG. 3B, FIG. 1B shows that the activity warm word ‘set timer’ may be added to the respective active set of warm words **112Ab** once the temperature of the smart oven is successfully pre-heated. Here, the AED **104** may receive an ambient context signal **440** from the smart oven that indicates that the smart oven is at a designated pre-heated temperature, thereby triggering the AED **104** to add the activity-based warm word ‘set timer’ to, and remove the activity-based warm word ‘pre-heat even’ from, the respective active set of warm words **112Ab** for the user **102b**. Notably, and as described in greater detail below, the addition and removal of the warm words ‘set timer’ and ‘pre-heat oven’ to and from the respective active set of warm words **112Ab** for the second user **102b** may trigger the warm word selector **400** to again execute the warm word arbitration routine **401** to determine whether to add the warm word

‘set timer’ to the final set of warm words **112F** enabled for detection by the AED **104**. In the example shown in FIG. **1B**, the AED **104** outputs a notification indicating once the smart oven is successfully preheated. For instance, the digital assistant **105** may generate synthesized speech **123** for audible output from the speaker **18** of the AED **104** that states, “Oven Reached 350-deg”. In additional examples, the digital assistant **105** may provide a notification to a user device **50** (e.g., smart phone) linked to user account of the identified user to inform the identified user **102b** (e.g., Jim) when the warm word ‘pre-heat oven’ has been added by the warm word selector **400** to the final set of warm words **112F** enabled for detection by the AED **104**.

(50) The GUI **300b** of FIG. **3B** also shows the option for the user to select other activity-based warm words for activities related to a smart doorbell and incoming calls. For instance, and as shown in FIG. **1B**, the smart doorbell may be designated activity-based warm words such as ‘Show Cam’, ‘Ignore’, and ‘Talk’ under visitor presence conditions such as when a visitor rings the smart doorbell and/or the smart doorbell detects the presence of a visitor proximate to the smart door bell. Under one or more of the visitor presence conditions, these activity-based warm words may be added to the respective active set of warm words **112A** for each user **102** that selected to include these activity-based warm words, e.g., via use of the warm word configurator screen presented by GUI **300b**. The smart doorbell may send an ambient context signal **440** (FIG. **4**) to the AED **104** that indicates the occurrence of one or more visitor presence conditions, whereby the AED **104** may sound a visitor notification **125** such as a doorbell chime for audible output from the speaker **18** of the AED **104**. The AED **104** may provide the visitor notification **125** as synthesized speech in addition to, or in lieu of, the doorbell chime. At the same time, the ambient context signal **440** may trigger the warm word selector **400** to execute the warm word arbitration routine **401** that determines to add the activity-based warm words ‘Show Cam’, ‘Ignore’, and ‘Talk’ to the final set of warm words **112F** enabled for detection by the AED **104**. Accordingly, and in the example of FIG. **1B**, any of the users **102a-c** can speak (without speaking a predefined hotword) ‘Show Cam’ to cause a screen in communication with the AED **104** to display image data of the visitor captured by a camera of the smart doorbell or proximate the smart doorbell, ‘Ignore’ to choose to dismiss the visitor notification (and optionally cause the smart doorbell to relay a pre-recorded message to the visitor), or ‘Talk’ to open a microphone in communication with the smart doorbell and the microphone **16** of the AED **104** (and/or microphone **52** of user device **50**) to provide intercom communication capabilities between the users **102** and the visitor.

(51) With continued reference to FIG. **3B**, the warm word configurator screen also shows a list of preferred warm words that the user **102b** may select for inclusion in the respective active set of warm words **112A** independent of activity. In the example shown, the user **102b** selects the warm words ‘Call [Contact]’, ‘Play Music’, and ‘Turn off lights’, ‘Turn on lights’ as preferred warm words to add the active set of warm words **112A**, **112Ab**. The list of preferred warm words may be prepopulated or based on commonly used voice commands learned over time that the digital assistant **105** promotes to warm words. Some of the preferred warm words may be custom words or phrases that the user assigns to a respective action (e.g., a routine) to be performed when spoken by the user. In the example shown, the warm words ‘What’s the weather’ and ‘Set Alarm’ are listed as preferred warm words but indicated as inactive warm words **112N** since the user **102b** did not select to add these warm words to the respective active list of warm words **112A**. The list of preferred warm words and activity-based warm words selected for inclusion in the respective active set of warm words **112A** may be saved in the respective warm word preferences **212** for each enrolled user **200**.

(52) Referring back to FIGS. **1A-1C** and FIG. **4**, the associated user device for a user **102** detected in the environment by the user detector **410** may push/broadcast the user’s **102** respective active set of warm words **112A** periodically and/or each time warm words are added or removed from the active set. Here, the user device may push/broadcast the respective active set of warm words **112A** from a respective user device **50** in communication with the AED **50**. Additionally or alternatively,

the warm word selector **400** may obtain the active set of warm words **112A** by accessing the warm word preferences **212** when the user detector **410** uniquely identifies the user as one of the enrolled users **200**. In some examples, the warm word selector **400** determines when warm words are added to active sets of warm words **112A** for different users responsive to receiving an ambient context signal **440** indicating a particular activity is currently in progress and the particular activity is assigned to one or more activity-based warm words conveyed in the warm word preferences **212**. The warm word selector **400** may obtain the active set of warm words **112** using other techniques such as prompting the detected user to provide his/her active set of warm words.

(53) Referring to FIG. 4, in some implementations, the warm word selector **400** executes the warm word arbitration routine **401** for enabling a final set of warm words **112F** for detection by the AED **104** based on the active set of warm words **112A** obtained for each user **102** among the multiple users **102** detected by the user detector **401**. The warm word selector **400** may execute the warm word arbitration routine **401** periodically during prescribed intervals and/or responsive to specific events/conditions. In some examples, the warm word arbitration routine **401** executes in response to determining addition of a new warm word to, or removal of one of the warm words from, the respective active set of warm words **112A** for one of the multiple different users present in the environment. To illustrate, FIG. 1B shows the first user **102a** (e.g., Barb) speaking a second utterance **106, 106b**, “Ok computer, stop music and start commute routine when I get in the car” in the vicinity of the AED **104**. The microphone **16** of the AED **104** receives the utterance **106b** and processes the audio data **502** to detect the hotword **110** using the hotword detector **108**, thereby causing the AED **104** to wake-up and perform speech recognition to recognize the command to stop the music and perform the action of starting Barb's commute routine once Barb is in her car. As a result of the command **118** causing the digital assistant **105** to stop the long-standing operation of playing back the music **122** from Barb's playlist, the music playback warm words of pause, next song, previous song, volume up, and volume down are now removed from the respective active set of warm words **112Aa** for Barb. In this example, the warm word selector **400** receives the respective active set of warm words **112Aa** for Barb which has been updated to now omit the music playback warm words, thereby triggering execution of the warm word arbitration routine **401** to potentially update the final set of warm words **112F** enabled for detection by the AED. Similarly, the addition of the activity-based warm words ‘Show Cam’, ‘Ignore’, and ‘Talk’ to the respective active set of warm words **112Aa-Ac** for each of the users **102a-c** may trigger the warm word selector **400** to execute the warm word arbitration routine **401**.

(54) In some examples, the warm word arbitration routine **401** executes in response to the user detector **410** detecting the presence of a new user within the environment of the AED **104**. Notably, since the AED obtains a respective active set of warm words **112A** for any new user detected, the warm word arbitration routine **401** executes to determine whether the final set of warm words **112F** should be updated to include any of the active warm words for the new user. By the same notion, updating the final set of warm words **112F** may include removing some warm words from the final set to make room (i.e., in terms of processing/memory capability and/or false acceptance thresholds) for any of the new active warm words for the new user.

(55) Additionally or alternatively, the warm word arbitration routine **401** may execute in response to the AED **104** determining that one of the previously detected users is no longer present within the environment of the AED **104**. For instance, FIG. 1C shows that the first user **102a** is no longer present in the environment. The user detector **410** may continuously update a list of users whom are currently detected in the environment of the AED **104**, thereby allowing the AED **104** to ascertain when a user is no longer present.

(56) In some implementations, the AED **104** determines a change in ambient context and the warm word arbitration routine **401** executes in response to determining the change in ambient context. In some examples, the warm word selector **400** receives the ambient context signal **440** from a source indicating the change in ambient context. For example, the smart doorbell may provide the ambient

context signal **440** indicating the occurrence of one or more visitor presence conditions and the oven could provide the ambient context signal **440** when the oven has reached the pre-heated temperature. An incoming call to one or more enrolled users **200** of the AED **104** could also serve as an ambient context signal **440** that causes the warm word arbitration routine **401** to potentially update the final set of warm words **112F** to include new warm words related to the incoming phone call event such as, without limitation, warm words such as ‘answer’ or ‘ignore’ that when spoken would cause the AED **104** to answer or ignore the incoming call.

(57) Execution of the warm word arbitration routine **401** may include obtaining enabled warm word constraints **430** and determining a number of warm words to enable in the final set of warm words **112F** for detection by the AED **104** based on the enabled warm word constraints **430**. The warm word constraints **430** may permit the warm word arbitration routine **401** to determine a warm word capacity for the AED **104** at any given time. The warm word constraints **430** may include at least one of memory and computing resource availability of the AED **104** for detection of warm words, computational requirements for enabling each warm word in the respective active set of warm words **112A** for each user of the multiple users present in the environment of the AED, an acceptable false accept rate tolerance, or an acceptable false reject rate tolerance. The acceptable false accept and reject rate tolerances could be based on global settings, device settings, or be user-defined on a per-user basis. In some examples, the acceptable false accept and reject rate tolerances can be dynamic where a fading of warm word detection sensitivity occurs when a user is less active or further from the device.

(58) The memory and computing resource availability may take into account whether the AED **104** is battery powered, and if so, the current capacity of the battery. The memory and computing resource availability may further take into account current warm words in the final set of warm words, account applications currently running on the AED **104** in addition to processing capabilities and storage capacity.

(59) The computation requirements may determine memory and processing resources required to execute a respective warm word model associated with each warm word in the active sets of warm words **112A**. For instance, the computational requirements could include warm word model size/parameters for each warm word as well as whether any of the warm words are speaker specific requiring performance of speaker id process **500**. Additionally or alternatively, the computational requirements may determine memory and processing resources required to execute the speech recognizer **116** in a low-power mode sufficient for only recognizing utterances of the warm words. Notably, a low acceptable false accept rate tolerance may require more processing of a respective warm word model for detecting a warm word than a higher acceptable false accept rate tolerance.

(60) In some implementations, execution of the warm word arbitration routine **401** includes ranking the warm words in the respective active set of warm words from highest priority to lowest priority for each corresponding user **102** of the multiple users **102** based on warm word prioritization signals **413** and enabling the final set of warm words **112F** for detection by the AED **104** based on the rankings of the warm words in the respective active sets of warm words **112A** for each corresponding user **102**. Here, the warm word prioritization signals **413** may include at least one of usage frequency by the corresponding user of each warm word in the respective active set of warm words **112A**, a current state of the AED **104**, ambient context of the AED **104**, or co-presence information indicating prior warm word usage and/or operations performed by the AED when the corresponding user was previously present in the environment with one or more combinations of other ones of the multiple users. The current state of the AED may include any long-standing operations the digital assistant is currently performing on the AED **104**. For instance, if the digital assistant is playing back music on a highest volume setting the prioritization signal **413** may cause the warm word arbitration routine **401** to rank the warm words ‘play’ and ‘volume up’ in the active set of warm words **112A** with a low priority since it is unlikely the user will speak these warm words. The ambient context may include activity recognition performed by the AED based on one

or more signals. For instance, FIG. 1A depicts the third user **102c** watching a movie with his/her headphones on and having enabled a do not disturb mode on his/her user device **50**. The routine **401** may determine a user context for the user **103c** based on image data **412** received from the camera **19** (or another camera) indicating the user is wearing headphones and looking away from the AED **104** and/or a communication signal received from the user device **50** (or headphones) indicating the “do not disturb” mode is enabled on the user device. If enabled, headphones may also communicate signals indicating the headphones are currently worn and playing back audio content. The user context for the user **102c** may additionally or alternatively be ascertained based on a signal received indicating a movie is in progress. As a result, the routine **401** may rank the respective active set of warm words **112A** for the user **103c** with a lower priority than the active sets of warm words **112Aa**, **112Ab** for the first and second users **102a**, **102b**. However, as shown in FIG. 1B, the third user **103c** may permit visitor alert notifications while the do not disturb mode is enabled such that the ambient context signals **440** indicating the visitor is at the doorbell causes the digital assistant **105** to output a visitor alert audibly from user's headphones and/or visually by displaying a graphic on the user device **50** or television the user **103c** is currently viewing to inform the user **102c** of the visitor. Thus, while the respective active set of warm words **112Ac** are ranked with a low priority under the conditions of FIG. 1A and not included (except for “Call [Contact]” which is also active for the other users **102a**, **102b**) in the final set of warm words, the visitor alert served to the third user **102c** in FIG. 1B may change the current user context for the user **102c** to now indicate that the third user **102c** may be interested in interacting with the digital assistant **105** to ascertain information about the visitor at the doorbell. As shown in FIG. 1B, the warm words ‘Show Cam’, ‘Ignore’, and ‘Talk’ are included and ranked highest in the active sets of warm words **112Aa-Ac** for all of the users **102a-c** and ultimately selected for inclusion in the final set of warm words **112F** enabled for detection by the AED **104**.

(61) The co-presence information may indicate how warm words are utilized by the corresponding user in the presence of other users. For instance, the first user **102a** and the second user **102b** may each often speak the warm word “play music” in the presence of one another, whereas the first user **102a** seldom speaks the warm word ‘play music’ in the presence of the third user **102c**.

(62) The arbitration routine **401** may select higher priority warm words from the ranked active sets of warm words for all of the users **102** for inclusion in the final set of warm words **112F** to enable for detection by the AED **104**. As mentioned previously, the number of warm words included in the final set of warm words **112F** is limited by the enabled warm word constraints **430**. In some examples, to optimize execution of the arbitration routine **401**, the arbitration routine **401** considers only warm words from the top-N warm words in the respective active set of warm words **112A** ranked for each user. The value of N may be fixed or variable for different users. In examples implementing values of N that are variable among the different sets of users, the routine **401** may determine a warm word affinity score for each corresponding user **102** among the multiple users and enable the final set of warm words based on the warm word affinity score. The warm word affinity score for each corresponding user **102** may be based on a frequency of warm word usage by the corresponding user **102** and/or frequency of interactions between the corresponding user **102** and the digital assistant **105**. Here, frequency of warm word usage may indicate how often the corresponding user **102** uses warm words when interacting with the digital assistant **105**, while the frequency of interactions may indicate how often the corresponding user **102** interacts with the digital assistant **105** generally. The frequency of warm word usage and/or interactions may be further constrained by a current duration that the presence of the corresponding user **102** is detected within the environment of the AED **104**, specific days of the week and/or times of day. For instance, a user may frequently ask “What's the weather” every morning around the same time. The warm word affinity score may additionally or alternatively be based on at least one of a duration of presence of the corresponding user **102** in the environment of the AED **104**, a proximity of the corresponding user **102** relative to the AED **104**, or a current user context. For instance, proximity

information **54** for a corresponding user may cause the arbitration routine **401** to assign a higher priority to warm words in the respective active sets of warm words for users closer to the AED **104** than other users since closer users may be more likely to interact with AEDs than users further away.

(63) In the examples of FIGS. **1A** and **1B** depicting the third user **103c** watching the movie with his/her headphones on and having enabled the do not disturb mode on his/her user device **50**, the warm word arbitration routine **401** may determine a lower affinity score for the third user **103c** than the other users **103a**, **103b** since the current context of the user **102c** indicates a low likelihood that the user **103c** will interact with the digital assistant **103** and the proximity information **58** indicates the third user **103c** is located relatively far away from the AED **104** compared to the other users. Moreover, the affinity score for the first user **102a** may be further boosted if the first user **102c** subsequently uses one of the music playback warm words after issuing the command **118** to “play my music playlist” in the first utterance **106a**. Similarly, the affinity score for the second user **102b** may be boosted if the user **102b** frequently makes use of the warm word “pre-heat oven” while viewing a recipe in the cooking application **107** as shown in FIG. **1A**. By the same notion, the ranking of the warm word ‘Set Timer’ in the respective active set of warm words is higher in FIG. **1B** after the oven reaches the preheat temperature since there is now a higher likelihood that the user **102b** will put a dish in the oven and set the timer for the duration the dish is to bake in the oven. In FIG. **1B**, the ‘preheat oven’ warm word is now ranked lowest in the respective active set of warm words **112Ab** for the second user **102c**.

(64) To illustrate how a combination of current user context and proximity affect the warm word affinity score for the third user **102b**, the current user context and proximity for the third user **102c** in FIG. **1A** associate the third user **102c** with a low affinity score while the event of the visitor notification **125** in FIG. **1B** associates the second user **102b** with a slightly higher affinity score. However, in FIG. **1C**, the proximity information **58** now indicates the third user **102c** has moved closer to the AED **104** and the current user context now indicates the do not disturb mode is disabled and the user is no longer wearing his/her headphones. As a result, the arbitration routine **401** may increase the affinity score for the third user **102c** since there is a higher likelihood of the user **102c** interacting with the digital assistant **105** compared to the example depicted in FIG. **1B** and especially the example depicted in FIG. **1A**.

(65) In addition to duration of presence, the arbitration routine **401** may further predict an expected likelihood of user presence in the future and determine the affinity score based on the expected likelihood of user presence. For instance, when the operation identifier **124** determines that the transcription output from the speech recognizer **116** for the second utterance **106b** includes the command **118** ‘ . . . stop music and start commute routine when I get in the car’, the arbitration routine **401** may predict that the presence of the user **102a** in the environment will likely end. As a result, the arbitration routine may start gradually reducing the affinity score for the first user **102a** until the first user **102a** is no longer detected as shown in FIG. **1C**.

(66) With continued reference to FIGS. **1A-1C** and **4**, the warm word arbitration routine **401** may boost the scores for warm words shared by two or more of the active sets of warm words **112A** since there is an increased likelihood of these warm words being spoken. These shared warm words may be ranked higher in the respective active sets of warm words **112A** so that they are not inadvertently omitted from the top-N active warm words as discussed above. Additionally, the associated warm word models for detecting these shared warm words among multiple users may be merged with one another to limit the number of similar warm word detection models executing on the AED **104**. Moreover, a warm word that is shared by different users may cause the digital assistant **105** to perform a different action depending on which user spoke the warm word **112**. For instance, in the example of FIG. **1C**, the second user **102b** speaking “play music” enabled for detection in the final set of warm words may cause the digital assistant **105** to stream music from a first music player indicated as a preferred music player for the second user **102b**. By contrast, the

third user **102c** speaking the same warm word “play music” may cause the digital assistant **105** to stream music from a different second music player preferred by the third user **102c**. Notably, the speaker identification process **500** of FIG. 5 may be performed on audio data **502** characterizing utterances of the warm word in order to uniquely identifier the user that spoke the warm word so that the appropriate action (e.g., choosing which music player to stream music) can be performed by the digital assistant **105**. In this example, warm word arbitration routine **401** identifies that the respective active sets of warm words **112Ab**, **112Ac** each include the warm word **112** “play music”. Rather than enable detection for two separate warm word models **330** for the warm word **112** “play music”, the warm word arbitration routine selects only one warm word model **330** for the warm word **112** “play music”. In some implementations, the warm word arbitration routine **401** determines that the AED **104** has sufficient capacity and enables execution of a higher quality (i.e., additional parameters, additional operations, decreased latency, and/or increased sensitivity) warm word model **330** for the warm word **112** “play music”. Optionally, the warm arbitration routine **401** determines the AED **104** has sufficient capacity and enabled execution of a different architecture such as moving from a warm word model **330** to an ASR model.

(67) Referring to FIG. 1C, the first user **102a** is no longer detected and the affinity score for the third user **102c** is further boosted due to the third user **102c** moving closer to the AED **104** and no longer wearing headphones. Additionally, the playback of the first user's **102a** music playlist has been stopped. As a result, the final set of warm words **112F** enabled for detection include the warm words “tell me a joke”, “set a timer”, “play music”, “call [contact]”, and “turn off lights”. The “tell me a joke” warm word may be speaker specific to the third user **102c** and selected from the respective active set of warm words **112Ac** for the third user **102c** while the “set a timer” warm word may be speaker specific to the second user **102b** and selected from the respective active set of warm words **112Ab** for the second user **102b**. The warm words “play music”, “call [contact]”, and “turn off lights” are shared warm words that may be spoken by either of the users **102b**, **102c** to cause the respective action specified by the warm word to be performed by the digital assistant **105** when detected by the AED **104** in streaming audio (e.g., using the appropriate warm word model **330** or speech recognizer **116**) without inclusion of a predetermined hotword (e.g., Ok Computer). In the example, the third user **102c** speaks a third utterance **106**, **106c** that includes a warm word **112** “tell me a joke” from the final set of warm words **112F** enabled for detection by the AED **104**. Without performing speech recognition on the captured audio, the AED **104** may apply the warm word models **330** for the final set of warm words **112F** to identify whether the utterance **106c** includes any of the warm words included in the final set of warm words **112F**. The AED **104** compares the audio data **502** that corresponds to the utterance **106c** to the enabled warm word models **330** that correspond to the warm words **112** “tell me a joke”, “set a timer”, “play music”, “call [contact]”, and “turn off lights” and determines that the warm word model **330** enabled for the warm word **112** “tell me a joke” detects the warm word **112** “tell me a joke” in the utterance **102c** without performing speech recognition on the audio data **402**. Most notably, the user speaks the utterance **106c** of the warm word **112** without prefixing the utterance **106c** with a predetermined hotword **110** such that the appropriate warm word model **330** detects the presence of the warm word **112** in the audio data and triggers the AED **104** to invoke the digital assistant **105** to perform the action (e.g., retrieve and playback a joke) specified by the warm word. In some examples, the speech recognizer **116** operates in a low-power mode for recognizing only the presence of warm words **112** in the final set of warm words **112F**. In these examples, the speech recognizer **116** may recognize when one of the warm words in the final set of warm words **112F** is spoken not in the presence of a predefined hotword and invoke the digital assistant to perform the respective action. In the example, the digital assistant **105** retrieves a joke from a search engine, on-device application, or other source and outputs the joke audibly as synthesized speech **129**. The digital assistant **105** may additionally or alternatively output the joke as a textual representation displayed on a screen **13** of the AED **104** or other screen in communication with the AED **104**.

(68) FIG. 6 includes a flowchart of an example arrangement of operations for a method **600** of enabling a final set of warm words **112F** for detection by an assistant-enabled device (AED) **104** when multiple users **102** are present in an environment of the AED **104**. The operations performed by the method **600** may be described with reference to FIGS. 1-5. As used throughout the present disclosure, the “environment” of the AED **104** refers to users within a proximity for interacting with the AED through speech and optionally other means. Accordingly, the “environment” may include a same room or even household as the AED, a vehicle cabin, an exterior of a vehicle, a doctor's office where the AED is located, a place of business where the AED is located, as well as within a proximity of the AED sufficient for a microphone of the AED to capture audio. The data processing hardware **12** of the AED **104** may execute instructions stored on the memory hardware **14** that cause the AED to perform the operations. Optionally, the server **130** may operate some or all of the operations.

(69) At operation **602**, the method **600** includes detecting a presence of multiple users **102** within the environment of the AED **104**. The AED **104** executes a digital assistant **105**. The AED **104** may execute multiple digital assistants simultaneously in some configurations. At operation **604**, for each user **102** of the multiple users **102**, the method also includes a respective active set of warm words **112A** that each specify a respective action for the digital assistant **105** to perform.

(70) At operation **606**, based on the respective active set of warm words for each user of the multiple users, the method **600** includes executing a warm word arbitration routine **401** to enable a final set of warm words **112F** for detection by the AED **104**. Here, the final set of warm words **112F** enabled for detection by the AED **104** include warm words **112** selected from the respective active set of warm words **112A** for at least one user **102** of the multiple users **102** detected within the environment of the AED **104**.

(71) At operation **608**, while the final set of warm words are enabled for detection by the AED **104**, the method includes receiving audio data **502** corresponding to an utterance **106** captured by the AED **104**, detecting a warm word **112** from the final set of warm words **112F** in the audio data **502**, and instructing the digital assistant to perform the respective action specified by the detected warm word.

(72) In some examples, the final set of warm words **112F** are enabled for detection by activating, for each warm word in the final set of warm words **112F**, a respective warm word model **330** to run on the AED. Here, the method uses the activated respective warm word model **330** to detect the warm word in the audio data **502** without performing speech recognition on the audio data **502**. More specifically, detecting the warm word in the audio data may include extracting audio features in the audio data **502**, generating a warm word confidence score by using the activated respective warm word model **330** to process the extracted audio features, and determining that the audio data corresponding to the utterance includes the warm word when the warm word confidence score satisfies a warm word confidence threshold. In some examples, the warm word confidence threshold may be ascertained for a corresponding enrolled user **200** by accessing the acceptable false accept rate tolerances and/or acceptable false reject rate tolerances stored in the warm word preferences **212** for the enrolled user **200**. In additional examples, when the final set of warm words **112F** includes two different warm words that are phonetically similar, the warm word confidence thresholds for detecting each of these warm words may be increased to reduce the propensity for false acceptances where a warm word model **330** wrongly detects a warm word in audio instead of a phonetically similar warm word that was actually spoken.

(73) FIG. 7 is a schematic view of an example computing device **700** that may be used to implement the systems and methods described in this document. The computing device **700** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or

claimed in this document.

(74) The computing device **700** includes a processor **710**, memory **720**, a storage device **730**, a high-speed interface/controller **740** connecting to the memory **720** and high-speed expansion ports **750**, and a low speed interface/controller **760** connecting to a low speed bus **770** and a storage device **730**. Each of the components **710**, **720**, **730**, **740**, **750**, and **760**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **710** (e.g., the data processing hardware **10**, **132** of FIG. **1**) can process instructions for execution within the computing device **700**, including instructions stored in the memory **720** or on the storage device **730** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as display **780** coupled to high speed interface **740**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **700** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

(75) The memory **720** stores information non-transitorily within the computing device **700**. The memory **720** (e.g., the memory hardware **12**, **134** of FIG. **1**) may be a computer-readable medium, a volatile memory unit(s), or non-volatile memory unit(s). The non-transitory memory **720** may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by the computing device **700**. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

(76) The storage device **730** is capable of providing mass storage for the computing device **700**. In some implementations, the storage device **730** is a computer-readable medium. In various different implementations, the storage device **730** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. In additional implementations, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **720**, the storage device **730**, or memory on processor **710**.

(77) The high speed controller **740** manages bandwidth-intensive operations for the computing device **700**, while the low speed controller **760** manages lower bandwidth-intensive operations. Such allocation of duties is exemplary only. In some implementations, the high-speed controller **740** is coupled to the memory **720**, the display **780** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **750**, which may accept various expansion cards (not shown). In some implementations, the low-speed controller **760** is coupled to the storage device **730** and a low-speed expansion port **790**. The low-speed expansion port **790**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

(78) The computing device **700** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **700a** or multiple times in a group of such servers **700a**, as a laptop computer **700b**, or as part of a rack server system **700c**.

(79) Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof.

These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(80) A software application (i.e., a software resource) may refer to computer software that causes a computing device to perform a task. In some examples, a software application may be referred to as an “application,” an “app,” or a “program.” Example applications include, but are not limited to, system diagnostic applications, system management applications, system maintenance applications, word processing applications, spreadsheet applications, messaging applications, media streaming applications, social networking applications, and gaming applications.

(81) The non-transitory memory may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by a computing device. The non-transitory memory may be volatile and/or non-volatile addressable semiconductor memory. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

(82) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

(83) The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(84) To provide for interaction with a user, one or more aspects of the disclosure can be

implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

(85) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A computer-implemented method when executed by data processing hardware causes the data processing hardware to perform operations comprising: detecting a presence of multiple users within an environment of an assistant-enabled device (AED), the AED executing a digital assistant; for each user of the multiple users, obtaining a respective active set of warm words that each specify a respective action for the digital assistant to perform; based on the respective active set of warm words for each user of the multiple users, executing a warm word arbitration routine to enable a final set of warm words for detection by the AED by: for each respective warm word in the respective active set of warm words for each user of the multiple users, determining memory and processing resources required to execute a respective warm word model on the AED that is associated with the respective warm word; identifying any shared warm words corresponding to warm words present in the respective active set of warm words for each of at least two of the multiple users; and determining a number of warm words to enable in the final set of warm words for detection by the AED based on; assigning, for inclusion in the final set of warm words, a higher priority to warm words identified as shared warm words; and the memory and processing resources determined for each of the warm words in the respective active set of warm words for each user of the multiple users, wherein the number of warm words in the final set of warm words enabled for detection by the AED comprise warm words selected from the respective active set of warm words for at least one user of the multiple users detected within the environment of the AED; and while the final set of warm words are enabled for detection by the AED: receiving audio data corresponding to an utterance captured by the AED; detecting, in the audio data, a warm word from the final set of warm words using the respective warm word model executing on the AED that is associated with the warm word without performing speech recognition on the audio data; and instructing the digital assistant to perform the respective action specified by the detected warm word.

2. The computer-implemented method of claim 1, wherein detecting the presence of the multiple users within the environment of the AED comprises detecting at least one of the multiple users is present within the environment based on proximity information of the AED relative to a user device associated with the at least one of the multiple users.

3. The computer-implemented method of claim 1, wherein detecting the presence of the multiple users within the environment of the AED comprises: receiving image data corresponding to a scene of the environment; and detecting the presence of at least one of the multiple users within the environment based on the image data.

4. The computer-implemented method of claim 1, wherein detecting the presence of the multiple users within the environment of the AED comprises detecting the presence of a corresponding user

within the environment of the AED based on: receiving voice data characterizing a voice query issued by the corresponding user and directed toward the digital assistant; performing speaker identification on the voice data to identify the corresponding user that issued the voice query; and determining that the identified corresponding user that issued the voice query is present within the environment of the AED.

5. The computer-implemented method of claim 4, wherein: the voice query issued by the corresponding user comprises a command that causes the digital assistant to perform a long-standing operation specified by the command; and obtaining the respective active set of warm words comprises adding, in response to the digital assistant performing the long-standing operation specified by the command, to the respective active set of warm words for the corresponding user that issued the voice query, one or more warm words specifying respective actions for controlling the long-standing operation.

6. The computer-implemented method of claim 1, wherein the operations further comprise: detecting the presence of a new user within the environment of the AED, wherein the warm word arbitration routine executes in response to detecting the presence of the new user within the environment.

7. The computer-implemented method of claim 1, wherein the operations further comprise: determining that one of the multiple users is no longer present within the environment of the AED, wherein the warm word arbitration routine executes in response to determining that the one of the multiple users is no longer present within the environment.

8. The computer-implemented method of claim 1, wherein the operations further comprise: determining addition of a new warm word to, or removal of one of the warm words from, the respective active set of warm words for one of the multiple users present in the environment of the AED, wherein the warm word arbitration routine executes in response to determining the addition of the new warm word to, or the removal of the one of the warm words from, the respective active set of warm words for the one of the multiple users present in the environment of the AED.

9. The computer-implemented method of claim 1, wherein the operations further comprise: determining a change in ambient context of the AED, wherein the warm word arbitration routine executes in response to determining the change in ambient context.

10. The computer-implemented method of claim 1, wherein executing the warm word arbitration routine further comprises: obtaining enabled warm word constraints, the enabled warm word constraints comprising at least one of: memory and computing resource availability on the AED for detection of warm words; an acceptable false accept rate tolerance; or an acceptable false reject rate tolerance; and, wherein determining the number of warm words to enable in the final set of warm words for detection by the AED is further based on the enabled warm word constraints.

11. The computer-implemented method of claim 1, wherein executing the warm word arbitration routine comprises: for each corresponding user of the multiple users, ranking the warm words in the respective active set of warm words from highest priority to lowest priority based on warm word prioritization signals, the warm word prioritization signals comprising at least one of: usage frequency by the corresponding user of each warm word in the respective active set of warm words; a current state of the AED; ambient context of the AED; or co-presence information indicating prior warm word usage and/or operations performed by the AED when the corresponding user was previously present in the environment with one or more combinations of other ones of the multiple users; and enabling the final set of warm words for detection by the AED based on the rankings of the warm words in the respective active set of warm words for each corresponding user of the multiple users.

12. The computer-implemented method of claim 1, wherein executing the warm word arbitration routine comprises: for each corresponding user of the multiple users, determining a warm word affinity score; and enabling the final set of warm words for detection by the AED based on the warm word affinity score determined for each corresponding user.

13. The computer-implemented method of claim 12, wherein the warm word affinity score determined for each corresponding user is based on at least one of: frequency of warm word usage by the corresponding user; frequency of interactions between the corresponding user and the digital assistant; duration of presence of the corresponding user in the environment of the AED; a proximity of the user relative to the AED; or a current user context.

14. The computer-implemented method of claim 1, wherein the operations further comprise: determining that the warm word detected in the audio data corresponding to the utterance captured by the user device comprises a speaker-specific warm word selected from the respective active set of warm words for each of one or more users among the multiple users such that the digital assistant only performs the respective action specified by the speaker-specific warm word when the speaker-specific warm word is spoken by any of the one or more users corresponding to the respective active set of warm words; and based on determining that the warm word detected in the audio data corresponding to the utterance captured by the user device includes a speaker-specific warm word, performing speaker verification on the audio data to determine that the utterance was spoken by one of the one or more users corresponding to the respective active set of warm words that the detected warm word was selected, wherein instructing the digital assistant to perform the respective action specified by the detected warm word is based on the speaker verification performed on the audio data.

15. The computer-implemented method of claim 1, wherein: the final set of warm words are enabled for detection by activating, for each warm word in the final set of warm words, the respective warm word model to run on the AED; and detecting, in the audio data, the warm word from the final set of warm words comprises detecting, using the activated respective warm word model, the warm word in the audio data without performing speech recognition on the audio data.

16. The computer-implemented method of claim 15, wherein detecting the warm word in the audio data comprises: extracting audio features of the audio data; generating, using the activated respective warm word model, a warm word confidence score by processing the extracted audio features; and determining that the audio data corresponding to the utterance includes the warm word when the warm word confidence score satisfies a warm word confidence threshold.

17. A system comprising: data processing hardware; and memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising: detecting a presence of multiple users within an environment of an assistant-enabled device (AED), the AED executing a digital assistant; for each user of the multiple users, obtaining a respective active set of warm words that each specify a respective action for the digital assistant to perform; based on the respective active set of warm words for each user of the multiple users, executing a warm word arbitration routine to enable a final set of warm words for detection by the AED by: for each respective warm word in the respective active set of warm words for each user of the multiple users, determining memory and processing resources required to execute a respective warm word model on the AED that is associated with the respective warm word; identifying any shared warm words corresponding to warm words present in the respective active set of warm words for each of at least two of the multiple users; and determining a number of warm words to enable in the final set of warm words for detection by the AED based on: assigning, for inclusion in the final set of warm words, a higher priority to warm words identified as shared warm words; and the memory and processing resources determined for each of the warm words in the respective active set of warm words for each user of the multiple users, wherein the number of warm words in the final set of warm words enabled for detection by the AED comprise warm words selected from the respective active set of warm words for at least one user of the multiple users detected within the environment of the AED; and while the final set of warm words are enabled for detection by the AED: receiving audio data corresponding to an utterance captured by the AED; detecting, in the audio data, a warm word from the final set of warm words using the respective warm word model

executing on the AED that is associated with the warm word without performing speech recognition on the audio data; and instructing the digital assistant to perform the respective action specified by the detected warm word.

18. The system of claim 17, wherein detecting the presence of the multiple users within the environment of the AED comprises detecting at least one of the multiple users is present within the environment based on proximity information of the AED relative to a user device associated with the at least one of the multiple users.

19. The system of claim 17, wherein detecting the presence of the multiple users within the environment of the AED comprises: receiving image data corresponding to a scene of the environment; and detecting the presence of at least one of the multiple users within the environment based on the image data.

20. The system of claim 17, wherein detecting the presence of the multiple users within the environment of the AED comprises detecting the presence of a corresponding user within the environment of the AED based on: receiving voice data characterizing a voice query issued by the corresponding user and directed toward the digital assistant; performing speaker identification on the voice data to identify the corresponding user that issued the voice query; and determining that the identified corresponding user that issued the voice query is present within the environment of the AED.

21. The system of claim 20, wherein: the voice query issued by the corresponding user comprises a command that causes the digital assistant to perform a long-standing operation specified by the command; and obtaining the respective active set of warm words comprises adding, in response to the digital assistant performing the long-standing operation specified by the command, to the respective active set of warm words for the corresponding user that issued the voice query, one or more warm words specifying respective actions for controlling the long-standing operation.

22. The system of claim 17, wherein the operations further comprise: detecting the presence of a new user within the environment of the AED, wherein the warm word arbitration routine executes in response to detecting the presence of the new user within the environment.

23. The system of claim 17, wherein the operations further comprise: determining that one of the multiple users is no longer present within the environment of the AED, wherein the warm word arbitration routine executes in response to determining that the one of the multiple users is no longer present within the environment.

24. The system of claim 17, wherein the operations further comprise: determining addition of a new warm word to, or removal of one of the warm words from, the respective active set of warm words for one of the multiple users present in the environment of the AED, wherein the warm word arbitration routine executes in response to determining the addition of the new warm word to, or the removal of the one of the warm words from, the respective active set of warm words for the one of the multiple users present in the environment of the AED.

25. The system of claim 17, wherein the operations further comprise: determining a change in ambient context of the AED, wherein the warm word arbitration routine executes in response to determining the change in ambient context.

26. The system of claim 17, wherein executing the warm word arbitration routine further comprises: obtaining enabled warm word constraints, the enabled warm word constraints comprising at least one of: memory and computing resource availability on the AED for detection of warm words; an acceptable false accept rate tolerance; or an acceptable false reject rate tolerance; and, wherein determining the number of warm words to enable in the final set of warm words for detection by the AED is further based on the enabled warm word constraints.

27. The system of claim 17, wherein executing the warm word arbitration routine comprises: for each corresponding user of the multiple users, ranking the warm words in the respective active set of warm words from highest priority to lowest priority based on warm word prioritization signals, the warm word prioritization signals comprising at least one of: usage frequency by the

corresponding user of each warm word in the respective active set of warm words; a current state of the AED; ambient context of the AED; or co-presence information indicating prior warm word usage and/or operations performed by the AED when the corresponding user was previously present in the environment with one or more combinations of other ones of the multiple users; and enabling the final set of warm words for detection by the AED based on the rankings of the warm words in the respective active set of warm words for each corresponding user of the multiple users.

28. The system of claim 17, wherein executing the warm word arbitration routine comprises: for each corresponding user of the multiple users, determining a warm word affinity score; and enabling the final set of warm words for detection by the AED based on the warm word affinity score determined for each corresponding user.

29. The system of claim 28, wherein the warm word affinity score determined for each corresponding user is based on at least one of: frequency of warm word usage by the corresponding user; frequency of interactions between the corresponding user and the digital assistant; duration of presence of the corresponding user in the environment of the AED; a proximity of the user relative to the AED; or a current user context.

30. The system of claim 17, wherein the operations further comprise: determining that the warm word detected in the audio data corresponding to the utterance captured by the user device comprises a speaker-specific warm word selected from the respective active set of warm words for each of one or more users among the multiple users such that the digital assistant only performs the respective action specified by the speaker-specific warm word when the speaker-specific warm word is spoken by any of the one or more users corresponding to the respective active set of warm words; and based on determining that the warm word detected in the audio data corresponding to the utterance captured by the user device includes a speaker-specific warm word, performing speaker verification on the audio data to determine that the utterance was spoken by one of the one or more users corresponding to the respective active set of warm words that the detected warm word was selected, wherein instructing the digital assistant to perform the respective action specified by the detected warm word is based on the speaker verification performed on the audio data.

31. The system of claim 17, wherein: the final set of warm words are enabled for detection by activating, for each warm word in the final set of warm words, the respective warm word model to run on the AED; and detecting, in the audio data, the warm word from the final set of warm words comprises detecting, using the activated respective warm word model, the warm word in the audio data without performing speech recognition on the audio data.

32. The system of claim 31, wherein detecting the warm word in the audio data comprises: extracting audio features of the audio data; generating, using the activated respective warm word model, a warm word confidence score by processing the extracted audio features; and determining that the audio data corresponding to the utterance includes the warm word when the warm word confidence score satisfies a warm word confidence threshold.
